Distance Measurement Using Ultrasonic Sensor

For

Sensor Lab mini Project

Third Year (Semester-VI) Bachelors in Information Technology, Autonomous Program

By

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Certificate

This is to certify that **Manorath Ital(19) Chirag Choudhary (10)** have completed the project report on the topic **Distance Measurement Using Ultrasonic Sensor** satisfactorily in partial fulfilment of the requirements for the award of Mini Project in sensor lab of third Year, (Semester-VI) in Information Technology under the guidance of **Mr. Abhishek Chaudhari** during the year 2024-2025 as prescribed by An Autonomous Institute Affiliated to University of Mumbai

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Abstract

In this project, we explore the implementation of a distance measurement system using the **HC-SR04 ultrasonic sensor**. The system is designed to accurately measure distances up to 4 meters with high precision, making it suitable for various industrial, robotic, and consumer-level applications. The ultrasonic sensor works by emitting high-frequency sound waves and measuring the time taken for the sound to reflect back from an object. Based on the time of flight, the distance to the object is calculated.

The **HC-SR04 ultrasonic sensor** operates on the principle of **ultrasonic echo** technology, where a sound pulse is sent out, and the time it takes for the pulse to return is used to determine the distance to the object. This technology is cost-effective, easy to implement, and widely used in robotic navigation, object detection, and obstacle avoidance applications.

In this system, the sensor is interfaced with an **Arduino Uno R3 microcontroller**. The Arduino is responsible for triggering the ultrasonic pulse, measuring the time of the echo, calculating the distance, and processing the data. The real-time distance data is displayed on a **16x2 LCD screen** using an I2C module, which provides an intuitive and compact interface for the user.

The entire system is powered using a **5V supply** from the Arduino, making it simple, portable, and energy-efficient. The I2C LCD module is used for easy communication and to minimize the number of connections, ensuring a neat and reliable setup.

This report covers the theoretical background of **ultrasonic sensing**, a detailed overview of the hardware and software components, the interfacing techniques, circuit design, step-by-step implementation, and analysis of the results. The project demonstrates a practical, low-cost solution for accurate distance measurement, with potential applications in **robotics**, **automation**, **obstacle avoidance**, and **educational projects**.

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Chapter 1

Introduction

1. Introduction

Accurate distance measurement plays a crucial role in several fields, such as robotics, industrial automation, and smart devices. Traditional distance measurement methods, like ultrasonic or infrared sensors, often struggle with performance issues in low-light or reflective environments. In contrast, laser sensors offer a reliable and precise alternative. This project utilizes the VL53L0X Time-of-Flight (ToF) sensor, which measures distances by calculating the time taken for the laser to reflect back from an object. The distance data is processed by an Arduino Uno and displayed on an LCD in real-time.

1.1 Objective

- To design a cost-effective and efficient laser-based distance measurement system.
- To provide real-time distance display using an Arduino microcontroller and an LCD.

1.2 Scope

- The system can be used in proximity sensors, obstacle detection, robotics, and automation.
- It works reliably in low-light conditions, leveraging laser sensing technology.

Chapter 2

Review of Literature

2.1 Survey

Application of Low-Cost Ultrasonic Sensor for Robot Environment Detection

This study investigates the application of ultrasonic sensors in robot environment detection. The research focuses on evaluating the accuracy, reliability, and versatility of ultrasonic sensors in various real-world conditions. Testing includes evaluating the sensor's performance on different surface materials (such as smooth, rough, and reflective surfaces) and under various lighting conditions, including both low-light and bright environments.

The ultrasonic sensor measures distances by emitting high-frequency sound waves and detecting the time taken for the sound to bounce back after hitting an object. The results show that the ultrasonic sensor is capable of providing reliable distance measurements up to a range of 4-5 meters, depending on environmental factors such as temperature and humidity.

The paper concludes that ultrasonic sensors are suitable for proximity sensing, object detection, and navigation tasks in robotic applications. Due to their robustness, relatively low cost, and ease of integration, ultrasonic sensors are ideal for a wide range of robotics tasks, such as obstacle avoidance and simple mapping, even in less controlled environments.

1] Real-time simulation of time-of-flight sensors

This paper discusses a real-time simulation model for Time-of-Flight (ToF) sensors, specifically the Photonic Mixing Device (PMD) type. These sensors are compact, low-cost, and widely applicable in robotics, automotive, and 3D imaging. The paper includes real-world effects such as motion blur, flying pixels, and deviation errors, with a physics-based approach ensuring realistic sensor behavior. The system uses GPU acceleration for interactive, real-time feedback and simulation control.

2] Study of arduino microcontroller board

This study explains the working principles of Arduino microcontrollers, highlighting their role in education and research, particularly in sensor-based projects. Arduino's ease of use and programming through the free Arduino IDE (Integrated Development Environment) make it an ideal platform for small-scale electronics projects. The study covers different types of Arduino boards and their applications, emphasizing Arduino's affordability and quick setup for beginner-level electronics.

2.2 Research Gap

Research Gap

1. **Limited Real-World Application Demonstrations**:
   While existing research on ultrasonic sensors often explores their technical specifications and performance in controlled settings, there is a significant gap in practical implementation examples for **low-cost distance measurement systems**. Specifically, the literature lacks projects where hobbyists, students, or DIY enthusiasts can build simple and effective systems based on ultrasonic sensors, providing an opportunity for real-world application demonstrations that are cost-effective and easy to replicate.

2. **Complex Environments Not Deeply Explored**:
   Most studies focus on testing ultrasonic sensors in laboratory conditions with ideal targets (e.g., flat surfaces, stable environments). However, the **accuracy of ultrasonic sensors** in **real-world environments** remains inadequately explored. Real-world conditions—such as fluctuating outdoor temperatures, non-flat or irregular surfaces, moving objects, or ambient noise—can significantly impact sensor performance. This gap presents a need for further studies examining how these sensors can maintain accurate readings under variable and challenging environmental conditions.

3. **Integration Simplicity for Beginners**:
   Despite the widespread use of **Arduino** in sensor-based projects, the research often overlooks **beginner-friendly implementations** of ultrasonic sensors. For example, there is limited emphasis on how to integrate **ultrasonic sensors** with **Arduino** and **LCD/OLED displays** in a way that is cost-effective, straightforward, and easy to understand for beginners. There's a need for more comprehensive guides and practical examples for novice users, demonstrating how to connect and use these components for simple, yet functional distance-measuring systems.

4. **Educational Usage**:
   While ultrasonic sensors have been widely studied in industrial applications and robotics, there is insufficient research focusing on how they can be **leveraged as educational tools** in learning environments. Ultrasonic sensors, combined with affordable platforms like Arduino, offer an excellent opportunity for **student projects** and **educational activities** in **electronics** and **embedded systems**. However, the existing literature does not address the potential of these sensors for enhancing **hands-on learning** or integrating them into **project-based learning environments**. The development of educational resources and step-by-step tutorials for using ultrasonic sensors in classroom or student projects is an area that needs further exploration.

Chapter 3

Project Description

2.1 Problem Definition

Accurate distance measurement is essential in applications like robotics, automation, and industrial monitoring. Traditional sensors, such as ultrasonic or infrared, often face challenges in providing precise measurements, especially in varying lighting or surface conditions. This project aims to develop a compact, cost-effective, and precise laser-based distance measurement system using the VL53L0X Time-of-Flight sensor and Arduino Uno R3. The system will measure distances up to 2 meters and display the output in real-time on an LCD using I2C communication.

The goal is to create a reliable and easy-to-build prototype that can be used in practical applications or extended into more advanced systems, such as robotics or wireless monitoring.

2.2 Steps involved

1. **Study of Sensor and I2C Communication**
   Understood the working principles of the VL53L0X ToF sensor and its communication via the I2C protocol.

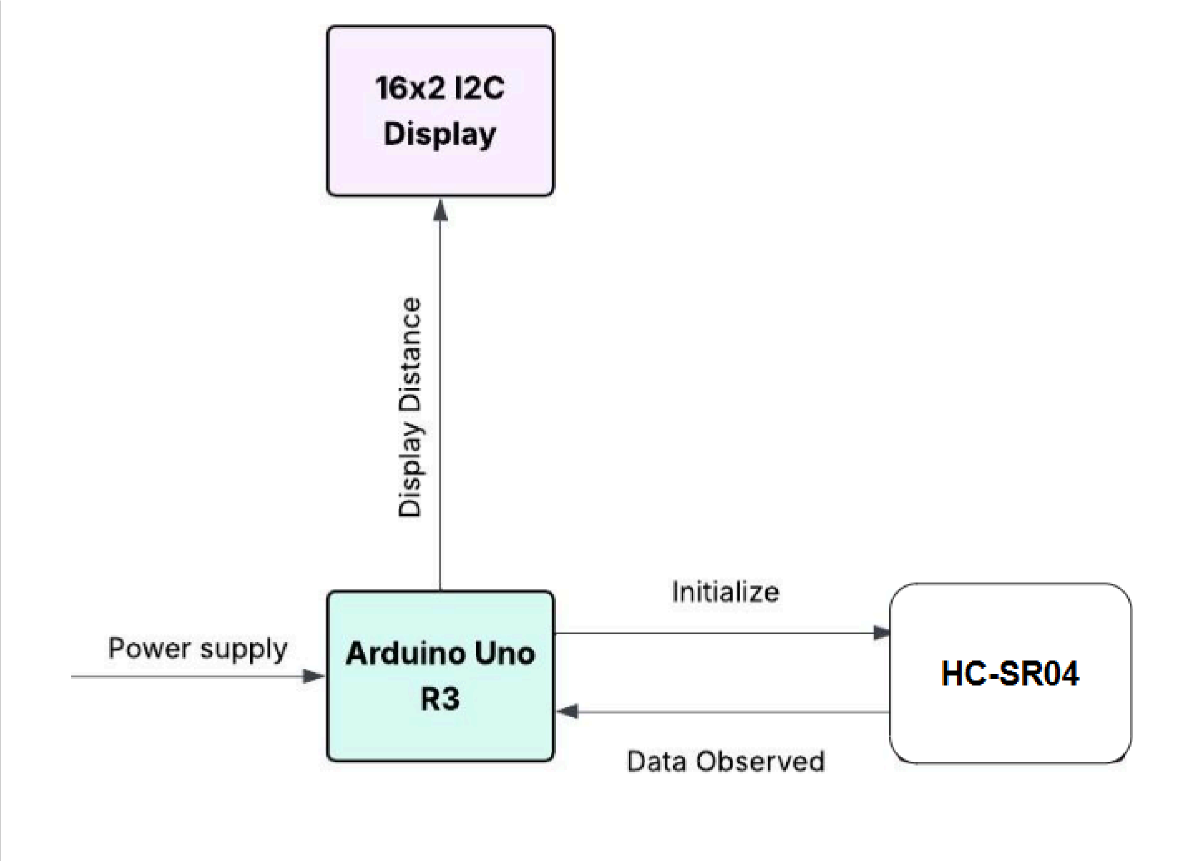
2. **Interfacing with Arduino Uno**
   Connected the sensor to the Arduino Uno R3 using the SDA and SCL pins and installed the necessary libraries.

3. **Displaying Values on LCD**
   Used a 16x2 LCD with an I2C module to display distance values in real-time.

4. **Circuit Design and Wiring**
   Built the circuit on a breadboard using jumper wires and ensured proper connections.

5. **Code Development and Testing**
   Wrote and uploaded the Arduino code for sensor reading and LCD display, then tested the setup for accuracy and reliability.

2.3 Block Diagram of proposed Project



2.4 Component Description

2.4.1 HC-SR04 Ultrasonic Sensor:

- Measures distance by emitting a high-frequency sound wave and measuring the time it takes for the sound to reflect back from an object.
- Range: Typically 2 cm to 400 cm.
- Operates at 5V DC.
- Works in most lighting conditions as it relies on sound, not light.

2.4.2 Arduino Uno R3

- Based on the ATmega328P microcontroller.

- Reads sensor data (pulse duration) to calculate the distance.
- Processes the data and sends it to the display or serial monitor.
- Powers the sensor and other connected modules.

2.4.3 LCD Display (16x2 or OLED)

- Displays real-time distance values.
- Uses an I2C interface for reduced wiring complexity.
- Connected to the Arduino via SDA (A4) and SCL (A5).

2.4.4 Jumper Wires & Breadboard

- Used for prototyping the circuit.
- Connects the Arduino, ultrasonic sensor, and display in a modular, easy-to-assemble manner.

2.4.5 Power Supply

- Arduino supplies 5V DC to the sensor and display modules.
- Ensures portability and simplicity in wiring.

2.5 Working of proposed project

2.5.1 Working

The core functionality of the project revolves around the operation of the **Time-of-Flight (ToF) distance sensor**, which is capable of measuring distances using sensor. The complete working process of the proposed system can be broken down into the following detailed steps:

Ultrasonic Pulse Emission:

The **HC-SR04 ultrasonic sensor** emits a high-frequency sound pulse through its **Trigger** pin.

This sound pulse travels through the air until it hits an object and then bounces back to the sensor.

Echo Detection:

The **Echo** pin of the sensor detects the reflected sound pulse after it hits the object.

The time between sending the pulse and receiving the echo is measured.

Time-to-Distance Calculation:

The Arduino calculates the distance by multiplying the time difference (measured in microseconds) by the speed of sound (approximately 343 meters per second) and dividing by 2 (for the round trip of the pulse).

Formula:
Distance=(Time×343)2\text{Distance} = \frac{(Time \times 343)}{2}Distance=2(Time×343)

The calculated distance is in centimeters or millimeters.

Communication with Arduino Uno:

The ultrasonic sensor's **Trigger** and **Echo** pins are connected to the Arduino Uno.

The Arduino controls the timing of the pulse emission and echo detection, calculates the distance, and sends the result to the LCD for display.

Display of Distance:

The measured distance is shown on a **16x2 LCD** (or OLED display), updated in real-time, reflecting the current measurement.

1. **Power Supply and Portability**
   The setup is powered through the **Arduino Uno**, which can be connected to a **USB cable** or a **battery adapter**, making it portable and ideal for use in mobile systems or field applications like **robots** or **drones**.

2. **Accuracy and Performance in Varying Conditions**
   The **HC-SR04 ultrasonic sensor** works well in a range of environments, unaffected by light conditions. However, its accuracy can be influenced by the **reflectivity of surfaces**. Smooth, solid surfaces provide more reliable measurements, while soft or irregular surfaces can lead to errors. The sensor is also immune to low-light or dark conditions, unlike optical sensors, and offers reliable performance outdoors.

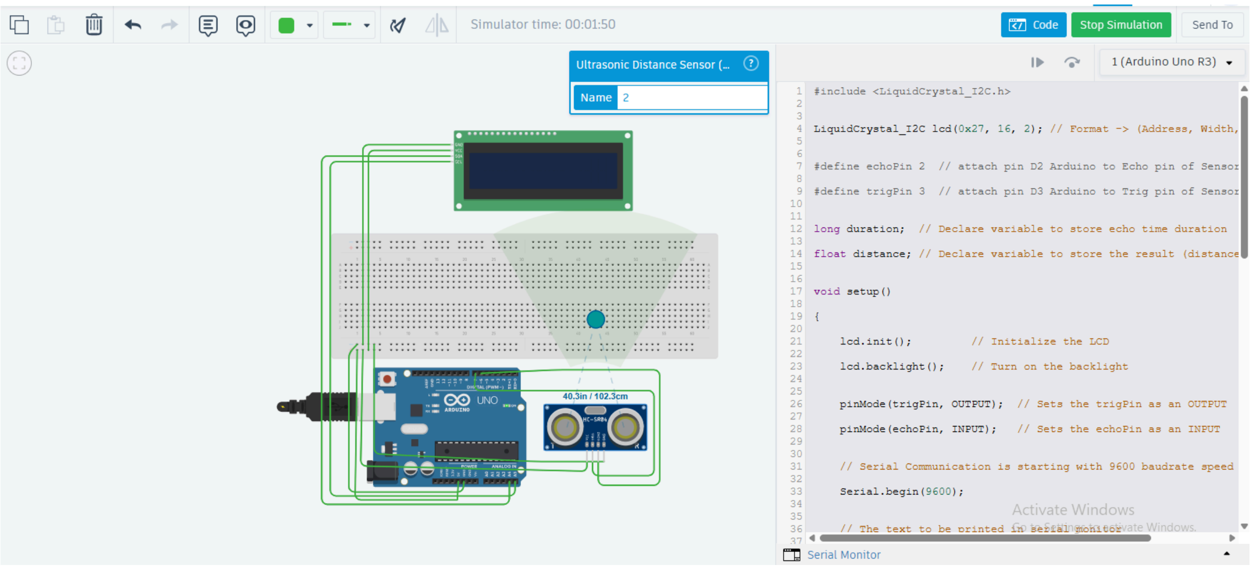
3.5.1 System Workflow

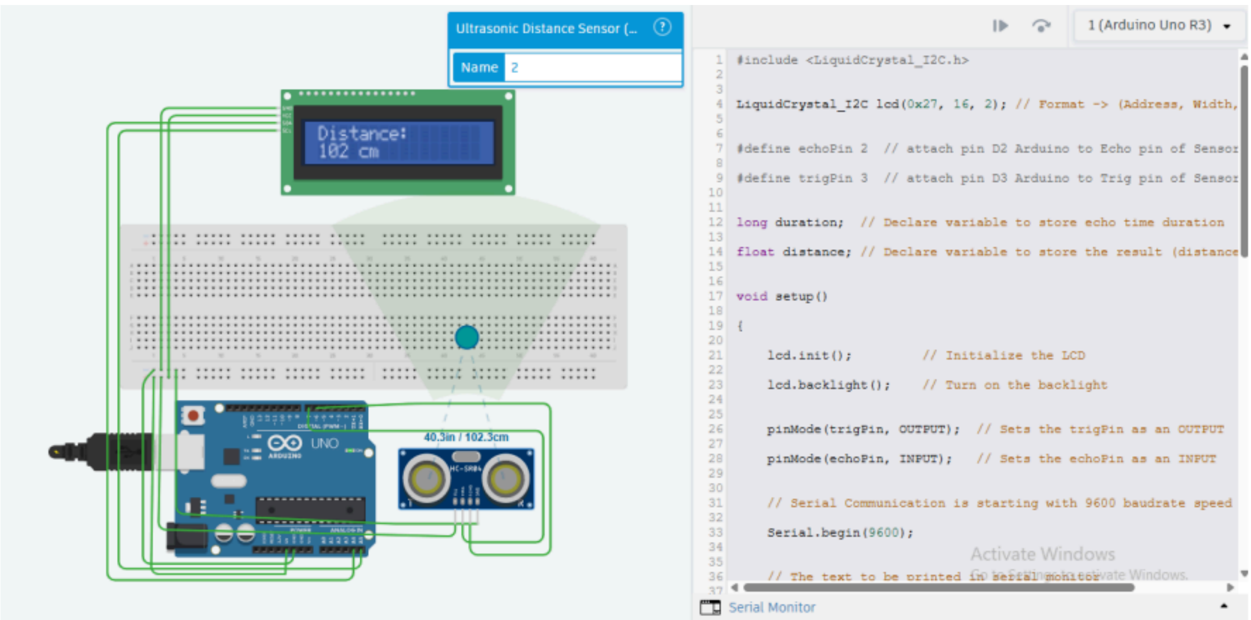
The system workflow begins with the initialization of the **HC-SR04 ultrasonic sensor** and the **LCD display** within the Arduino setup code. Once powered on, the Arduino configures the necessary pins for the ultrasonic sensor and ensures that the LCD display is ready for operation. In the main loop of the program, the Arduino continuously sends a trigger pulse to the ultrasonic sensor and waits for the echo pulse to measure the distance. These distance values are then processed and immediately sent to the LCD display for output. The display is updated in each cycle to show the latest distance measurement. Additionally, the Arduino code includes logic to handle potential measurement errors, such as out-of-range readings or temporary signal issues, by displaying error messages or retrying the measurement to ensure reliable performance.

Chapter 4

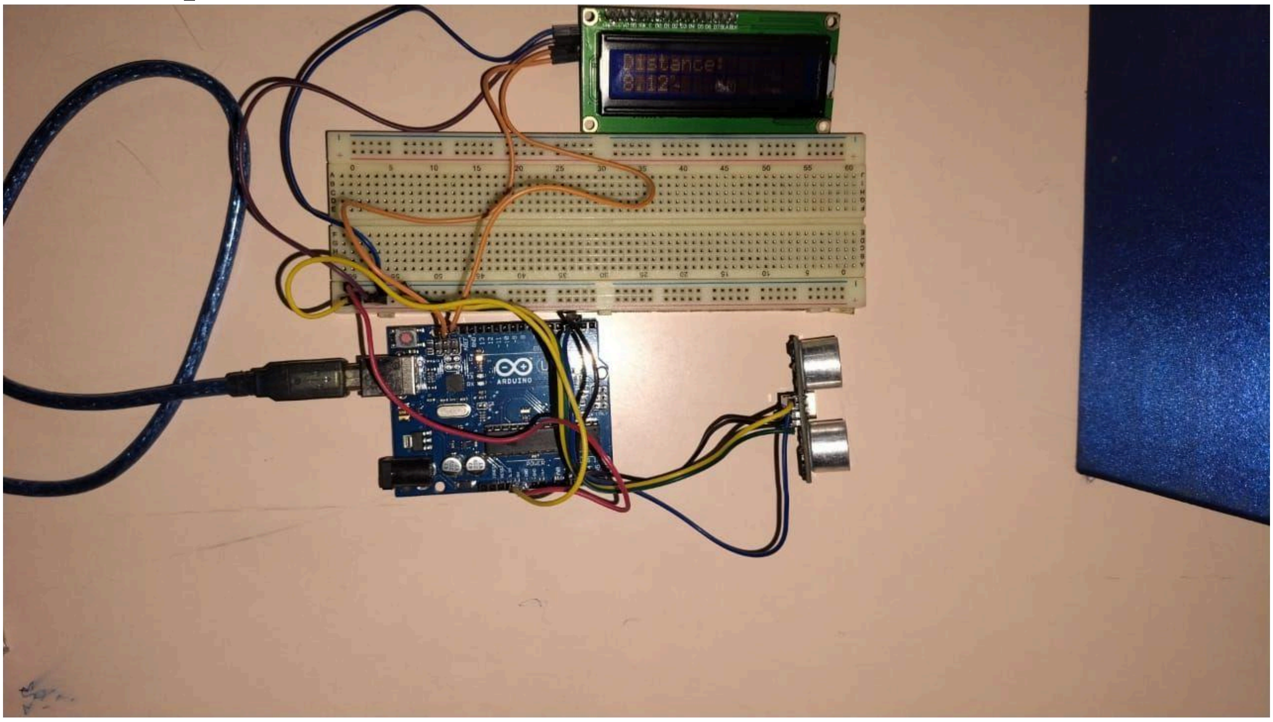
Implementation

4.1 Simulation





4.2 Actual Implementation



4.3 Hardware

4.3.1 Circuit Connection

- **Ultrasonic Sensor (HC-SR04):**
    - **VCC → 5V on Arduino**
    - **GND → GND on Arduino**
    - **Trig (Trigger Pin) → Pin 9 on Arduino (or any other available digital pin)**
    - **Echo (Echo Pin) → Pin 10 on Arduino (or any other available digital pin)**
- **LCD (I2C)**
    - **VCC → 5V on Arduino**
    - **GND → GND on Arduino**
    - **SDA → A4 on Arduino**
    - **SCL → A5 on Arduino**

4.3.2 Code:

```
#include <LiquidCrystal_I2C.h>

LiquidCrystal_I2C lcd(0x27, 16, 2); // Format -> (Address, Width, Height)

#define echoPin 2  // attach pin D2 Arduino to Echo pin of Sensor module
#define trigPin 3  // attach pin D3 Arduino to Trig pin of Sensor module

long duration;  // Declare variable to store echo time duration
float distance; // Declare variable to store the result (distance) as a decimal

void setup()
{
   lcd.init();        // Initialize the LCD
   lcd.backlight();    // Turn on the backlight

   pinMode(trigPin, OUTPUT);  // Sets the trigPin as an OUTPUT
   pinMode(echoPin, INPUT);   // Sets the echoPin as an INPUT

   // Serial Communication is starting with 9600 baudrate speed
   Serial.begin(9600);

   // The text to be printed in serial monitor
   Serial.println("Distance measurement using Arduino Uno");
   delay(500);
}

void loop()
{
   digitalWrite(trigPin, LOW);
   delayMicroseconds(2);
   digitalWrite(trigPin, HIGH);
   delayMicroseconds(10);
   digitalWrite(trigPin, LOW);

   duration = pulseIn(echoPin, HIGH);
   distance = duration * 0.0344 / 2.0; // Ensure floating point division

   // Print the result to the serial monitor
   Serial.print("Distance: ");
   Serial.print(distance, 2); // Print with 2 decimal places
   Serial.println(" cm");

   // Display the result on the LCD
```

```
    lcd.clear();             // Clear the display buffer
    lcd.setCursor(0, 0);     // Set cursor for "Distance:" (Column, Row)
    lcd.print("Distance:");  // Print "Distance:" at (0, 0)

    lcd.setCursor(0, 1);     // Set cursor for output value (0, 1)
    lcd.print(distance, 2);  // Print the distance with 2 decimal places

    lcd.setCursor(8, 1);     // Move cursor to (8, 1)
    lcd.print("cm");         // Print "cm" at (8, 1)

    delay(2000);
}
```

4.4 Software (Flowchart/ Algorithms)

The software is the core part of the distance measurement system and is developed using the Arduino Integrated Development Environment (IDE). Arduino IDE is an open-source platform that supports programming in C/C++ and is widely used for microcontroller-based applications. In this project, Arduino IDE is used to write, compile, and upload the code to the Arduino Uno R3.

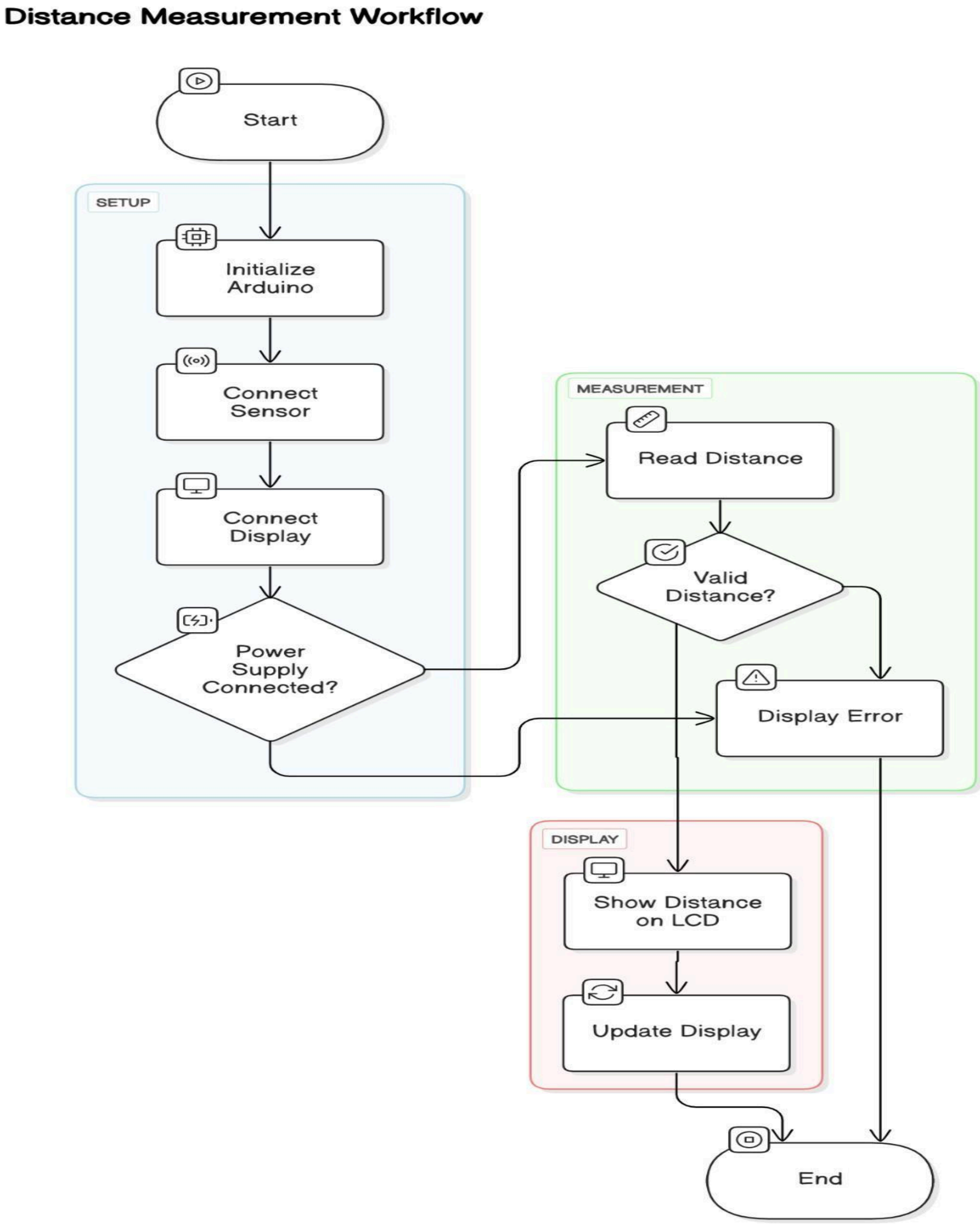
4.4.1 Software Implementation Process:

1. **Installing the Required Libraries:**
    - For the **LCD display (16x2 with I2C)**, we use the **LiquidCrystal_I2C** library to interface with the LCD.
    - For the **Ultrasonic sensor (HC-SR04)**, no external library is needed, as the `pulseIn()` function in Arduino can handle the echo time measurement.
        -
2. **Writing the Code:**
    - The Arduino sketch begins with including necessary libraries and defining sensor and LCD objects.
    - In the `setup()` function, the sensor and LCD are initialized.
    - In the `loop()` function, the sensor continuously reads the distance, and the result is printed on the LCD screen.
3. **Uploading to Arduino Uno:**
    - The code is compiled and uploaded to the Arduino board via a USB connection using the Upload button in the Arduino IDE.
    - The **Serial Monitor** can be used for debugging and checking values during development.

4.4.2 Algorithm:

- Start the program.
- Include necessary sensor and LCD libraries.
- Initialize the sensor and LCD in the setup() function.
- Initialize the LCD and turn on the backlight.
- Set the Trig pin as OUTPUT and Echo pin as INPUT for the ultrasonic sensor.
- Begin continuous distance measurement in the loop().
- Trigger the ultrasonic sensor by setting the Trig pin HIGH for 10 microseconds.
- Measure the echo time using `pulseIn()` function.
- Read distance from ultrasonic sensor.
- Calculate the distance using the formula:
  `distance = (duration * 0.0344) / 2.0`.
- Convert and format the result for display.
- Prepare the result for output on the LCD display.
- Print the distance on the LCD.
- Display the distance on the LCD screen with the label "Distance:".
- Delay slightly before the next reading for stability.
- Add a small delay (e.g., 2000 ms) to ensure stable measurements.
- Repeat indefinitely.

4.4.3 Flowchart



Chapter 5

Results and Conclusion

The distance measurement system using the **Ultrasonic Sensor (HC-SR04)** was successfully designed and implemented. The system demonstrated reliable and consistent performance in real-world testing conditions.

The ultrasonic sensor provided accurate distance readings up to 4 meters, with precision within a few centimeters. The readings remained stable across different surfaces and environments, showcasing the sensor's robustness. The use of digital communication via the **Trig** and **Echo** pins allowed seamless integration with the Arduino Uno R3, ensuring fast and efficient data transmission.

The measured distance values were displayed in real-time on a **16x2 LCD**, which updated dynamically with each reading cycle. This allowed users to observe immediate feedback during testing, making the system user-friendly and interactive.

Key observations from testing include:

- High accuracy in distance measurement with minimal fluctuation.
- Real-time display on LCD worked efficiently without noticeable lag.
- Stable performance even in varied environmental conditions.

The overall system was compact, lightweight, and cost-effective, making it ideal for small-scale, educational, and beginner-level projects.

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